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does vodafone uk support camara silent carrier handshake api?

13:26

Yes, Vodafone UK supports the underlying technology known as the CAMARA **Number Verify** API, which performs silent network-based mobile number verification (often referred to in the industry via the functional concept of a silent carrier/SIM-based handshake).  Vodafone Developer Mar... +1

CAMARA API Support on Vodafone UK

- **Number Verify 2.0:** Vodafone recently rolled out Number Verify 2.0 globally and in the UK, adhering to international [CAMARA standards](#). 
- **Silent Verification:** The API performs silent SIM-based checks via the carrier data connection rather than traditional SMS one-time passwords (OTPs).  Vodafone Developer Marke...
- **Industry Collaboration:** Vodafone UK collaborates alongside EE, Virgin Media O2, and Three on standardized [CAMARA network APIs](#) for anti-fraud and identity verification.  Telefónica Open Gateway +1

If you are looking to integrate this into an application, would you like details on accessing the Vodafone Developer Marketplace or specifics on endpoint implementation for **Number Verify 2.0**?

AI responses may include mistakes. [Learn more](#)



Ask anything



yes

13:27

To integrate Vodafone UK's CAMARA **Number Verify** API (v2.0) into your application, you must access it through Vodafone's centralized developer portal or partner aggregators.

Here is the technical blueprint to get started.

1. Accessing the Gateway

- **Vodafone Developer:** Register an account on the [Vodafone Developer Marketplace](#).
- **GSMA Open Gateway:** You can also access this API via authorized aggregators like Vonage, Infobip, or Twilio, who provide a single SDK for all UK carriers.

2. How the Silent Handshake Works

The API works on a three-way trust model:



Step A: Authentication (OAuth 2.0)

You must first request a client credentials token from the Vodafone Identity provider:

http

POST /oauth2/v1/token HTTP/1.1

Host: ://vodafone.com

Content-Type: application/x-www-form-urlencoded

grant_type=client_credentials&client_id=YOUR_ID&client_secret=YOUR_SEC

Use code with caution.



Step B: The Number Verify Call

Using the bearer token, make a `POST` request to the standardized CAMARA endpoint. **Crucial requirement:** The user must be browsing on mobile data (not Wi-Fi) for the silent handshake to validate the IP address automatically.

http

POST /camara/number-verification/v2/verify HTTP/1.1

Host: ://vodafone.com

Authorization: Bearer YOUR_ACCESS_TOKEN

Content-Type: application/json

```
{  
  "phoneNumber": "+447700900077"  
}
```

Use code with caution.



```
{  
  "devicePhoneNumberVerified": true  
}
```

Use code with caution.



Would you like assistance writing the **backend implementation code** (e.g., in Node.js or Python), or do you need the specific **fallback workflows** for when users are connected to Wi-Fi?

AI responses may include mistakes. [Learn more](#)



need the specific fallback workflows for when users are connected to Wi-Fi

13:30

Vodafone UK's rollout of **Number Verify 2.0** fundamentally changes how Wi-Fi environments are handled. In older legacy CAMARA implementations (v1.x), users were forced to drop down to a cellular connection or manual SMS because verification relied strictly on IP/Header enrichment over mobile data.  Vodafone Developer Ma... +2

With the latest architecture, there are two primary integration workflows to handle users on Wi-Fi.  Vodafone

Workflow 1: The Native Native OS / TS.43 Solution (Preferred)

Vodafone's Number Verify 2.0 natively integrates with the device operating system using the international **GSMA TS.43 Entitlement Server** standard



framework.  Ericsson +1

3. **SIM Verification:** The device operating system securely communicates with the Vodafone TS.43 entitlement server via an encrypted background channel.  Vodafone Developer Mar... +1
4. **Token Generation:** Vodafone verifies the active SIM profiles on the device, securely generates an operator token, and passes it back to your app.  LinkedIn · Mobile Ecosys... +1
5. **API Match:** Your backend passes this token to the `/verify` endpoint to securely complete the handshake.  LinkedIn · Mobile Ecosyste...

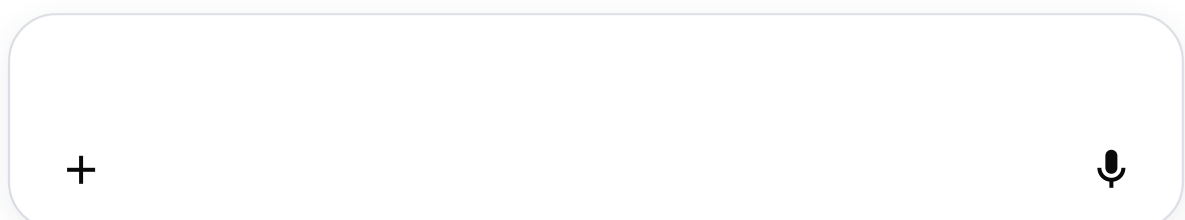
Workflow 2: Automated Network Transition (For Legacy/Web Clients)

If you are operating a web-based journey or a device that lacks native TS.43 OS support, you must configure a **Network Switchover**

Fallback:  bxbucket.blob.core.wind... +1

1. **Identify Wi-Fi:** Your application checks the active connection type.  Atlassian
2. **Temporary Cellular Wake:** For native apps (iOS/Android), the application triggers a temporary network transition. The app forces the authentication API request to route strictly over the cellular radio interface (using `ConnectivityManager` on Android or `NWPathMonitor` on iOS) while keeping the user's primary browsing on Wi-Fi.  bxbucket.blob.core.windo...
3. **Silent Verification:** The API request travels over Vodafone's mobile data network, triggers the standard cellular IP handshake, and validates the phone number.  Vodafone
4. **Return:** The app tears down the temporary cellular request path once verified.

Workflow 3: True Fallback Channels (For No-Cellular-Signal Situations)



- GSMA GSMA +1

Fi?

Learn more
.....



smartphones, tablets, and wearables rather than being restricted to specific phone models.  U2opia +3

The breakdown of device and operating system support includes:

1. Apple Ecosystem (iOS & watchOS)

- **Supported Devices:** iPhone (iPhone XS/XR and newer), iPad (Cellular models), and Apple Watch (Cellular models).
- **OS Implementation:** Apple uses a proprietary entitlement client framework that seamlessly interoperates with carrier [GSMA TS.43 Entitlement Servers](#). It handles silent tasks like Wi-Fi Calling provisioning, eSIM activation, and background network token generation.  Ericsson +2

2. Google Android Ecosystem (Android 12 and newer)

- **Supported Devices:** Devices from major manufacturers including Samsung Galaxy, Google Pixel, Motorola, and Nothing. 
- **OS Implementation:** Google officially integrated the [IMS Service Entitlement Feature](#) natively into the Android Open Source Project (AOSP) core beginning with **Android 12**.  Android Open Source Proje...
- **OEM Customisation:** While Google provides the standard TS.43 client package, manufacturers like Samsung choose to optimize or build upon it within their own skins (e.g., One UI) to handle multi-device eSIM and wearable syncing.  GitHub

Important Developer Restriction

While the hardware and operating systems fully support TS.43 communication with Vodafone, the native OS-level TS.43 APIs require **privileged/carrier-grade access**.  GitHub

An unprivileged third-party app cannot call the OS TS.43 engine directly. Instead, third-party apps leverage the standard **CAMARA Number Verify API** web endpoints. The carrier's network infrastructure and partner gateway aggregators (like Vonage or Prove) manage the backend bridging between your API request and the device's TS.43 entitlement state



AI responses may include mistakes. [Learn more](#)



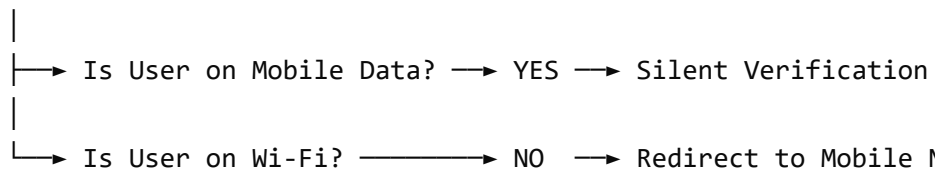
web-based service

13:34

Because you are building a **web-based service**, your environment operates under strict web-sandbox security controls. Unlike native mobile apps, web browsers (such as Chrome, Safari, or Edge) **cannot** access native TS.43 OS entitlement hooks, nor can they programmatically force a background network switch from Wi-Fi to cellular data. [🔗](#)



[User on Web Browser]



1. The Detection and Routing Strategy

When a user arrives at your web-based authentication or checkout page, you must determine their network path:

- **If browsing via Vodafone Mobile Data:** The browser naturally routes its HTTP requests through Vodafone's network gateways. The CAMARA API will automatically detect the user's header/IP, perform the silent handshake, and return a match instantly.
- **If browsing via Wi-Fi:** The browser routes traffic through a fixed broadband ISP (e.g., BT, Virgin Media). Because the traffic does not touch Vodafone's mobile gateway, the standard CAMARA `/verify` endpoint will return an error (typically an IP mismatch or unresolved network error). [🔗](#)

2. The Recommended Web Fallback Workflows

To maintain a high conversion rate on the web without compromising security, you should implement a tiered fallback system:

Option A: The "Temporary Wi-Fi Off" Prompt (Highest Conversion)

If your system detects the user is on Wi-Fi, display a short, user-friendly UI prompt before initiating the check.

1. **The Prompt:** "For secure, instant verification, please **turn off Wi-Fi** for a moment and tap continue."
2. **The Handshake:** Once the user toggles Wi-Fi off, their device switches to Vodafone mobile data.



1. Your backend requests a specialized verification URL from the gateway aggregator.
2. Your web app redirects the user's browser briefly to a specific Vodafone network URL.
3. If the user's phone has a cellular radio active in the background, the network attempts to ping and capture the SIM signature via a cellular data loopback, even while Wi-Fi is active.
4. The user is redirected back to your web app with a verification token. [↗](#)

Option C: Instant SMS OTP Fallback (The Safety Net)

If the user refuses to turn off Wi-Fi, or if Option B fails to establish a cellular loopback within a tight timeout (e.g., 3 seconds), your backend must immediately fall back to a traditional web verification method: [↗](#)

- **Passkeys (WebAuthn):** If the user has a registered device, prompt for a browser-native biometric check (FaceID/TouchID/Windows Hello).
- **SMS OTP:** Dynamically render a standard 6-digit text code field to complete the login journey.

Would you like to see a sample **frontend Javascript layout** showing how to present the "Turn off Wi-Fi" user prompt, or do you need information on which **UK web aggregators** provide the best cross-carrier redirect APIs?

AI responses may include mistakes. [Learn more](#)



what does vodafone uk provide?

13:37

The [Vodafone Developer Marketplace](#) provides a comprehensive suite of enterprise-grade, CAMARA-aligned Open Gateway Network APIs tailored for identity verification, fraud prevention, and network intelligence.  GitHub +1

The primary developer services and APIs provided by Vodafone UK include:

1. Identity & Anti-Fraud APIs (CAMARA Standardised)

- **Number Verification 2.0:** The next-generation [Number Verify API](#) that utilizes TS.43 entitlement server handshakes to validate a user's phone number silently over both mobile data and Wi-Fi connections.  mobileecosystemforum.... +1
- **SIM Swap:** Detects the exact timing of a user's last SIM card change to prevent account takeover and intercept fraud during high-risk actions like bank transfers.  Telefónica Open Gateway +1
- **Age Verification:** Cross-references network subscriber data to confirm if a user meets legal age-restricted thresholds without forcing them to upload physical ID documents.  GSMA +2
- **SIM Match:** Validates provided customer registration details (such as name and address) directly against official carrier billing



- **Brand Registration:** Allows enterprises to attach a verified, human-readable brand name string to their outbound corporate phone calls, which displays dynamically on the recipient's screen.
- **Verified Caller:** Authenticates the legitimacy of an incoming call to protect users from spoofing, spam, and voice-phishing attacks.
- **Call Forwarding Check:** Detects whether active conditional or unconditional call forwarding is enabled on a target mobile number.  www.telcotitans.com +4

3. Device & Connectivity Intelligence APIs

- **Device Roaming Status:** Identifies if a device is connected to a foreign network and logs the country it is currently located in for global fraud scoring.  www.telcotitans.com +1
- **Device Status (Connectivity):** Returns real-time metrics confirming whether a specific handset is actively connected to the cellular network and ready to receive data or SMS.  www.telcotitans.com
- **Device Information:** Provides hardware-level signatures and IMEI details associated with the active end-user device.  www.telcotitans.com
- **Quality on Demand (QoS):** Allows applications to request latency or bandwidth prioritization over the 5G network for high-demand tasks like live streaming or telemetry.  [Telefónica Open Gateway](https://telefonicaopengateway.com) +1

4. Developer Environments & Tooling

- **Vodafone CAMARA Sandbox:** A free, risk-free testing sandbox backed by the Vodafone API Lab, letting developers prototype RESTful network queries against mock data before moving to a live production environment.
- **Aggregator Ecosystem Accessibility:** In addition to the native developer portal, Vodafone provides cross-carrier availability for its API portfolio via global platforms like the Nokia Network as Code (NaC) platform, Vonage, and Twilio.  [Vodafone Developer Ma...](#) +4

Are you interested in exploring the documentation for a **specific identity**





how does Number Verification 2.0 work with roaming

13:48

Vodafone UK's **Number Verify 2.0** natively supports international roaming. It maintains the silent, background verification handshake even when a user leaves the UK and connects to a foreign partner network.  Telefónica Open Gateway +1

The technical architecture shifts to handle roaming under the hood using specific mechanisms.



- **The Handshake:** Because the mobile data session terminates at Vodafone UK's core Home Subscriber Server (HSS), Vodafone can still map the data session to the user's UK mobile number. [🔗](#)
- **The Result:** The CAMARA API backend queries Vodafone UK's infrastructure directly, meaning the silent verification executes exactly the same way it would if the user were physically in London. [📰 Tech Times](#)

2. How TS.43 Overcomes Roaming Wi-Fi Blocks

In older legacy versions, being on a hotel Wi-Fi network while roaming broke the process completely. With **Number Verify 2.0**, the TS.43 Entitlement Server solves this: [🔴 Vodafone Developer Mar... +1](#)

1. The device operating system connects to the hotel Wi-Fi.
2. The browser or app triggers the verification request.
3. The device's native OS establishes an encrypted, secure background channel directly back to Vodafone UK's TS.43 entitlement server.
4. The active roaming SIM generates an un-phishable, cryptographic hardware token.
5. This token bypasses the local Wi-Fi provider and authenticates directly with Vodafone, validating the user securely. [🔴 Vodafone Developer Ma... +2](#)

3. Developer Implementation Impact (Cross-Border Interoperability)

Because Number Verify 2.0 is strictly built on the standardized **GSMA/CAMARA Open Gateway open framework**, you do not need to alter your web application logic to accommodate roaming users. [🔴 Mobile World Live +1](#)

- Your backend continues to send the standardized `POST /camara/number-verification/v2/verify` API payload.
- The aggregation gateways route the lookup internationally to Vodafone UK automatically based on the user's country code prefix (`+44`). [🌐 mobileecosystemforum.com](#)



have no cellular pathway. It will default completely to local Wi-Fi. In this specific scenario, the fallback workflows (such as a temporary network switchover or SMS fallback) become mandatory.  Vodafone +2

- **Network Latency Timeouts:** Because a roaming data session must tunnel out of the visited country back to Vodafone UK, latency increases. Ensure your backend API timeout limits are configured to be slightly more forgiving (e.g., 5 seconds instead of 2 seconds) to avoid dropping legitimate roaming authentication requests.

If you are mapping out the edge cases for your web service, would you like guidance on incorporating the **CAMARA Device Roaming Status API** alongside Number Verify to flag high-risk country mismatches?  GSMA

AI responses may include mistakes. [Learn more](#)



No, you cannot use the CAMARA Number Verification API to retrieve or "look up" an unknown user's phone number [the functional concept of a silent phone number lookup].

The API is strictly designed for **identity validation**, not identity discovery. [↗](#)

How the Data Flow Restrictions Work

- **Input Requirement:** Your application's backend must already know and provide the user's phone number (in international format, e.g., `+447700900077`) inside the API request payload.
- **The Response Limit:** The Vodafone network will only return a simple true/false match (`devicePhoneNumberVerified: true/false`). It will never reveal, leak, or transmit the actual phone number over the wire. [↗](#)
- **Privacy by Design:** This strict limitation is built directly into the global CAMARA security architecture to comply with GDPR, UK Data Protection Acts, and privacy standards. It prevents malicious websites from silently harvesting user phone numbers in the background. [↗](#)

How to Handle User Identification on the Web

Because the API cannot give you the number, your web service must collect it first: [↗](#)

1. **User Input:** Prompt the user to type their mobile number into a standard input field on your website.
2. **API Verification:** Pass that inputted number to the Vodafone CAMARA API to silently verify if the SIM card inside their phone matches what they just typed.
3. **Session Binding:** Once the network returns `true` , you can safely bind that verified phone number to the active user session. [↗](#)

If you are looking for a way to log users in completely seamlessly without making them type a number, would you like to explore **Passkeys (WebAuthn)** or **OAuth-based Mobile Connect** frameworks instead?



what does GET device-phone-number do in the vodafone implementation

14:42

The `GET /device-phone-number` endpoint handles **secure retrieval of the active phone number directly from the user's SIM card and network profile**.  Vodafone Developer Marke...

While the `POST /verify` endpoint only confirms or denies a number you provide (`true` / `false`), the `GET` endpoint securely sends the phone number back to your system.  GSMA +1

How `GET /device-phone-number` Operates

- **Zero Manual Input:** It identifies the phone number currently associated



- **3-Legged Authentication:** It requires a "3-legged" OAuth 2.0 access token. This means Vodafone verifies the user's live network session (the first leg), gets their consent (the second leg), and authenticates your server credentials (the third leg) before the number is returned. [GSMA +2](#)

The Core Difference: Verify vs. Retrieve

Feature	POST /verify	GET /device-phone-number
Primary Goal	Authentication & Fraud Prevention	Autofill & Frictionless Sign-up
Your Payload	You must send the phone number	No phone number sent
Network Response	true or false match	Full phone number string (e.g., +4477...)
User Friction	Completely silent (No popups)	Shows an explicit network consent screen

Common Use Case

This endpoint is most commonly used for **one-click onboarding or registrations**. A user lands on your registration page, taps "Sign up instantly with Vodafone," grants permission, and your system immediately retrieves their official billing phone number to create their account. [Camara Project +2](#)

Would you like to see how the **OAuth 2.0 3-legged authorization code flow** is structured to obtain the user consent token needed for this specific `GET` request?

AI responses may include mistakes. [Learn more](#)



+

wait. we are going to build a complete system after the info gathering exercise

14:44

That makes complete sense. Pausing to gather all the technical parameters up front ensures your architectural design is rock solid before writing a single line of production code.

Since this is a web-based service targeting Vodafone UK users, let's catalog the specific structural constraints we have identified so far:

- **API Purpose:** Your system will use `POST /verify` for silent validation (anti-fraud/session binding) or `GET /device-phone-number` for seamless onboarding (user registration).
- **Network Modes:** The system must gracefully handle users browsing natively on Vodafone mobile data (instant match) versus users on Wi-Fi (requiring a "turn off Wi-Fi" UX prompt, a dynamic carrier redirect loop, or an SMS OTP fallback).
- **Roaming Logic:** The implementation will natively support international



to-machine validation, and a 3-legged user consent flow to retrieve actual numbers.

To finish rounding out your architecture blueprint, are there any other network APIs or operational factors you need to evaluate? For instance, we can look into **SIM Swap tracking** for transaction security, or review how to handle cross-carrier support for **EE, Three, and O2** using the same code base. Let me know what information you need next.

AI responses may include mistakes. [Learn more](#)



we creating o proof of concept for vodafone only. sim swap tracking not needed yet

14:47

Got it. Keeping the Proof of Concept (PoC) laser-focused strictly on Vodafone UK ensures you can validate the core silent verification mechanics cleanly without any architectural bloat.

To complete the parameters for your informational blueprint, here are the final three pillars specific to a Vodafone-only web implementation:

1. The Environment Split

Vodafone divides its platform access into two tiers, which you will need to scope out during the upcoming build phase:

- **The Vodafone Developer Sandbox:** Uses mock responses to test your web logic, token acquisition, and fallback paths. You do not need a live SIM card or network routing for this stage.
- **The Live Production Gateway:** Requires your server to connect over TLS



Because this PoC is restricted strictly to Vodafone UK, your web application must protect its infrastructure by checking the network before invoking the API. If an EE or O2 user accidentally hits your PoC URL, the CAMARA API cannot verify them. Your web gateway should look at the incoming request's IP routing or use a basic front-end check to ensure only Vodafone clients trigger the token flows.

3. Session State & Token Lifespans

- **2-Legged Tokens (/verify):** These client credential tokens typically last 1 hour. Your backend system should cache them locally rather than requesting a new token for every individual web verification attempt.
- **3-Legged Tokens (/device-phone-number):** These consent-based tokens are short-lived (usually expiring within minutes) and single-use. They must be consumed immediately to retrieve the number and cannot be reused across different web sessions.

With all the foundational mechanics, roaming logic, Wi-Fi web fallbacks, and token architectures now catalogued, what **next steps** would you like to take to wrap up this gathering phase? We can dive into mapping the **step-by-step sequential data flow diagrams**, or transition directly into building the core **system architecture layout**.

AI responses may include mistakes. [Learn more](#)



lets start with a core architecture layout

14:49





1. Client Layer (User Web Browser)

- ## 2. Application Layer (Your Backend)

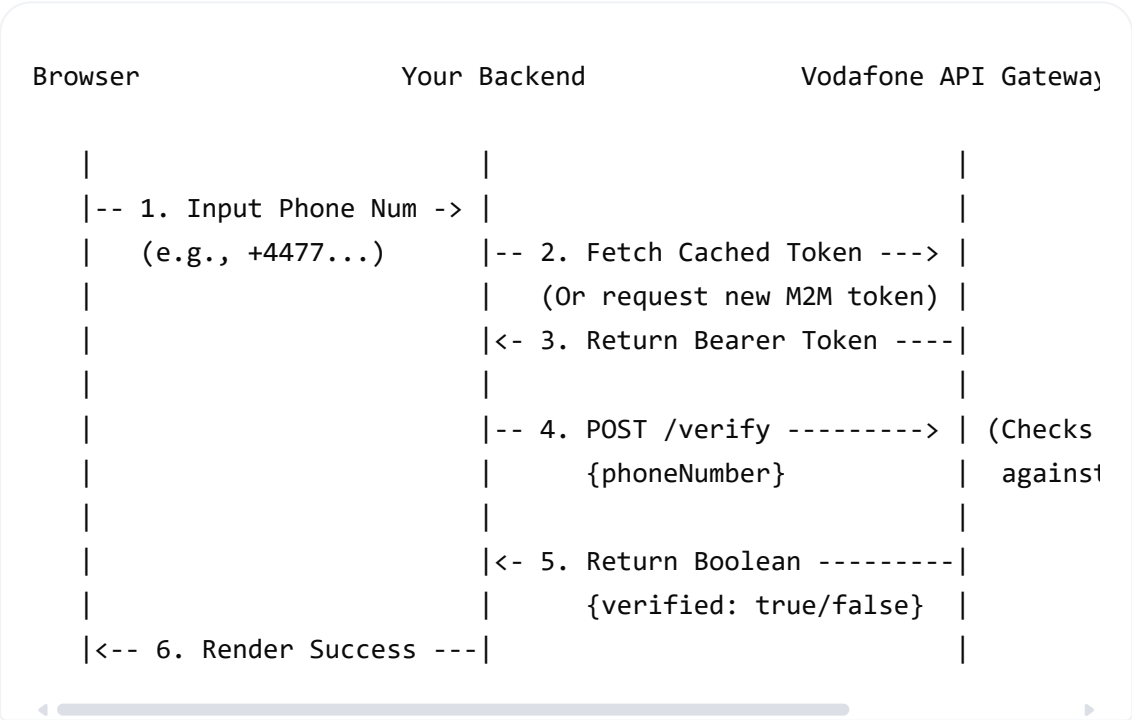
- **CAMARA API Gateway Engine:** Exposes the standardized `POST /verify` and `GET /device-phone-number` endpoints. It cross-references incoming traffic with real-time network states.
- **TS.43 Service Entitlement Server:** Manages cryptographic, tokenized hardware authentication states directly with the underlying mobile operating system platform.

Sequential Step-by-Step Data Flow

To map out exactly how components communicate during a live transaction, we track the execution sequence for our two core target use cases.

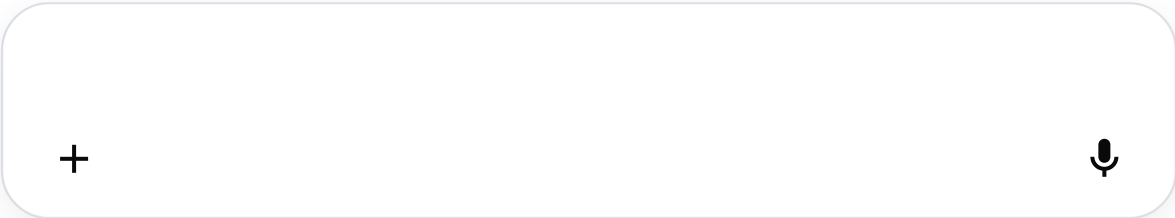
Flow A: Silent Number Verification (`POST /verify`)

This sequence executes when validating an identity without displaying manual authorization popups to the user.



Flow B: One-Click Registration (`GET /device-phone-number`)

This sequence executes when your application needs to securely retrieve the user's phone number straight from the SIM card.



```
|<-- 3. Redirect to Consent URL -----|
|                                     |
|-- 4. User Grants Permission On-Screen ----->| (SIM si{
|                                     |
|<-- 5. Redirect back to App with Auth Code -----|
|                                     |
|-- 6. Send Auth Code -->|                                     |
|                                     |-- 7. Exchange Code for Token>|
|                                     |<- 8. Return Access Token ----|
|                                     |
|                                     |-- 9. GET /device-phone-number|
|                                     |<- 10. Return Phone Number ---|
|<-- 11. Creation Success |                                     |
```

Now that the structural layout and step-by-step transaction streams are mapped out, should we move directly into drafting the **technical data validation schemas** for the API payloads, or would you prefer to sketch the **frontend logic state machine** that coordinates the Wi-Fi fallback behavior?

AI responses may include mistakes. [Learn more](#)



Please confirm that vodafone Number Verification v2.1.0 can work with both mobile data and wifi Also we want t...

14:54

Yes, I can confirm that the **Vodafone Number Verification v2.1.0 API** natively supports operation over **both mobile data and Wi-Fi connections**. This capability is powered by its integration with the underlying GSMA TS.43 Entitlement Server framework, which allows a secure, background cryptographic handshake with the device's SIM card even when



and fully supported by the [Vodafone Developer Portal's](#) endpoints: 

1. **Step 1 (GET /number-verification/v2/device-phone-number)**: This pulls the authenticated phone number string directly from the active SIM card profile after the user completes the OAuth 3-legged consent pop-up.  [developers.opengateway.t...](#)
2. **Step 2 (POST /number-verification/v2/verify)**: Your backend then captures that number string and passes it into the body payload of the verify call ({ "phoneNumber": "+4477..." }) to re-confirm that the user's active token is explicitly bound to that exact MSISDN line identity.  Vodafone Developer Mar... +1

Design Considerations for the Web PoC

While this architecture works seamlessly for a PoC, please note the following operational traits for your upcoming system build:

- **User Consent Layer**: Calling the GET endpoint requires an explicit browser-based OAuth user consent redirect loop. It cannot run "silently" in the background without user interaction.  [mobileecosystemforum.com](#)
- **Redundant Logic**: In a full production system, using POST immediately after a successful GET is often redundant. The GET request *already* authenticates and extracts the trusted number directly from the carrier network. Most architectures deploy GET for the sign-up workflow (autofill onboarding) and reserve POST for silent, unprompted login/checkout step-up verifications where the system already stores the user's number on file.  Camara Project +2

Now that we have confirmed the environment capabilities, would you like to proceed with writing out the **mock API payload requests and responses** for this multi-step flow, or should we look at the **OAuth 3-legged authentication token exchange code** your backend will need to process?

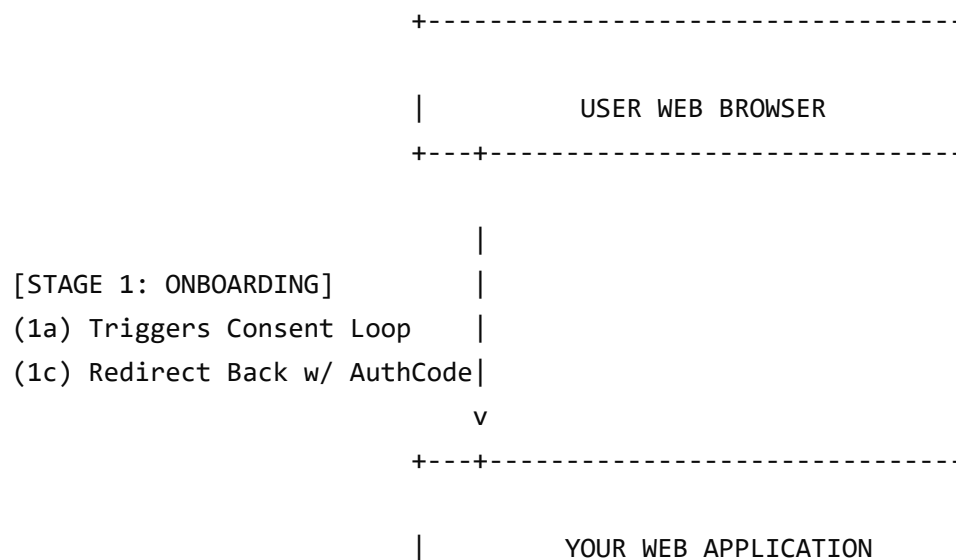
AI responses may include mistakes. [Learn more](#)

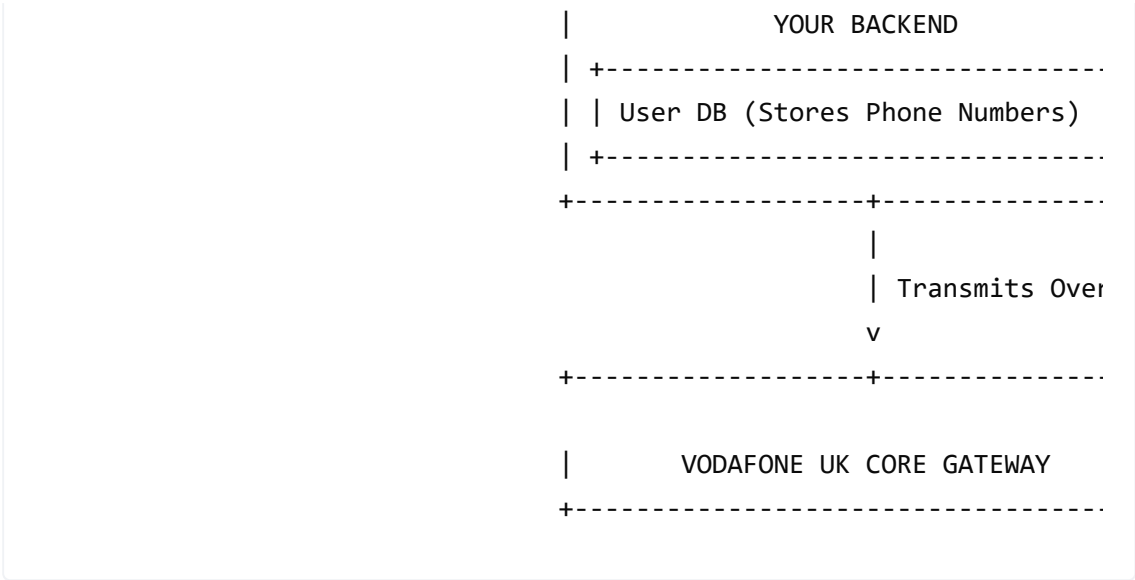
👍 🗑️ 🔄 📄 🗑️



i need a new architecture layout for user consent when initially onboarding using GET to retrieve phone number... ▾

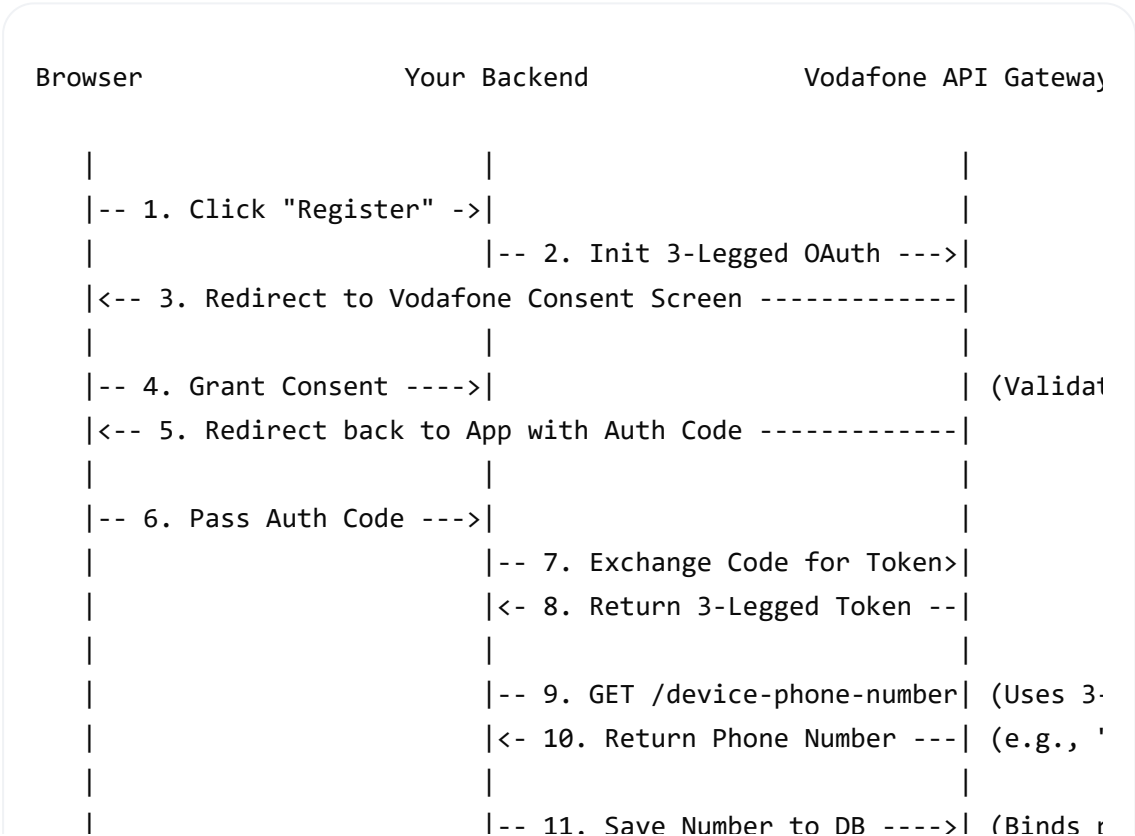
15:03



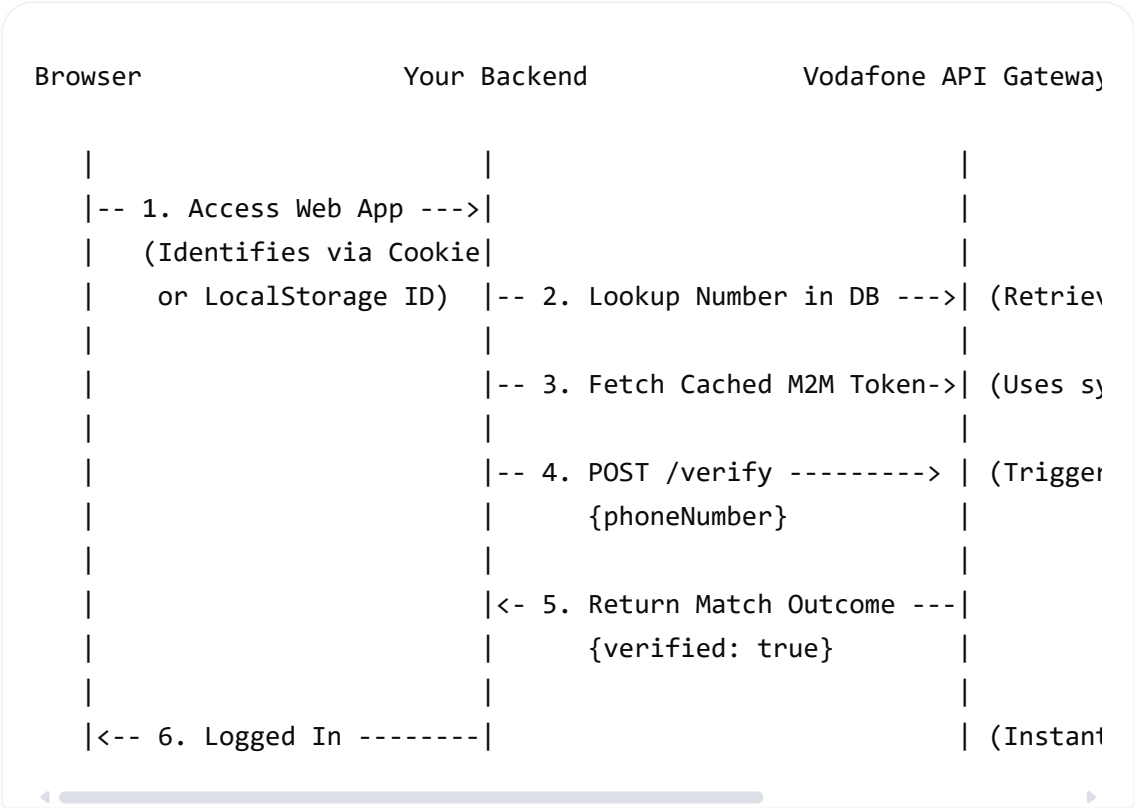


Stage 1: Initial Onboarding Workflow (Retrieving via GET)

This sequence executes only **once** per user profile during registration. It utilizes a **3-legged OAuth flow** that requires explicit browser consent to extract the network phone number.



This sequence executes seamlessly **every time** the user logs back into your service. It uses a cached **2-legged machine-to-machine (M2M) token** and runs silently in the background over Wi-Fi or mobile data without any popups.



Operational Architecture Summary

- **User Friction:** Onboarding (Stage 1) involves a clear, mandatory network permission dialog screen. Returning access (Stage 2) is completely frictionless and silent.
- **Database Dependency:** Your backend database serves as the link between the two stages. It stores the phone number extracted during the first stage so it can be automatically injected into the verification call during the second stage.
- **Token Separation:** Stage 1 uses a personalized, temporary 3-legged authorization token. Stage 2 relies entirely on your application's persistent, reusable 2-legged infrastructure credentials.



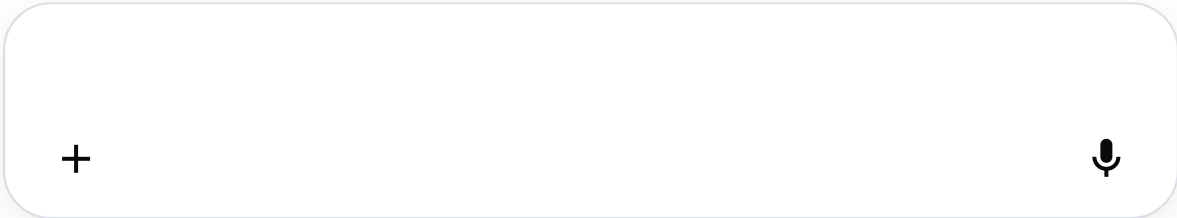
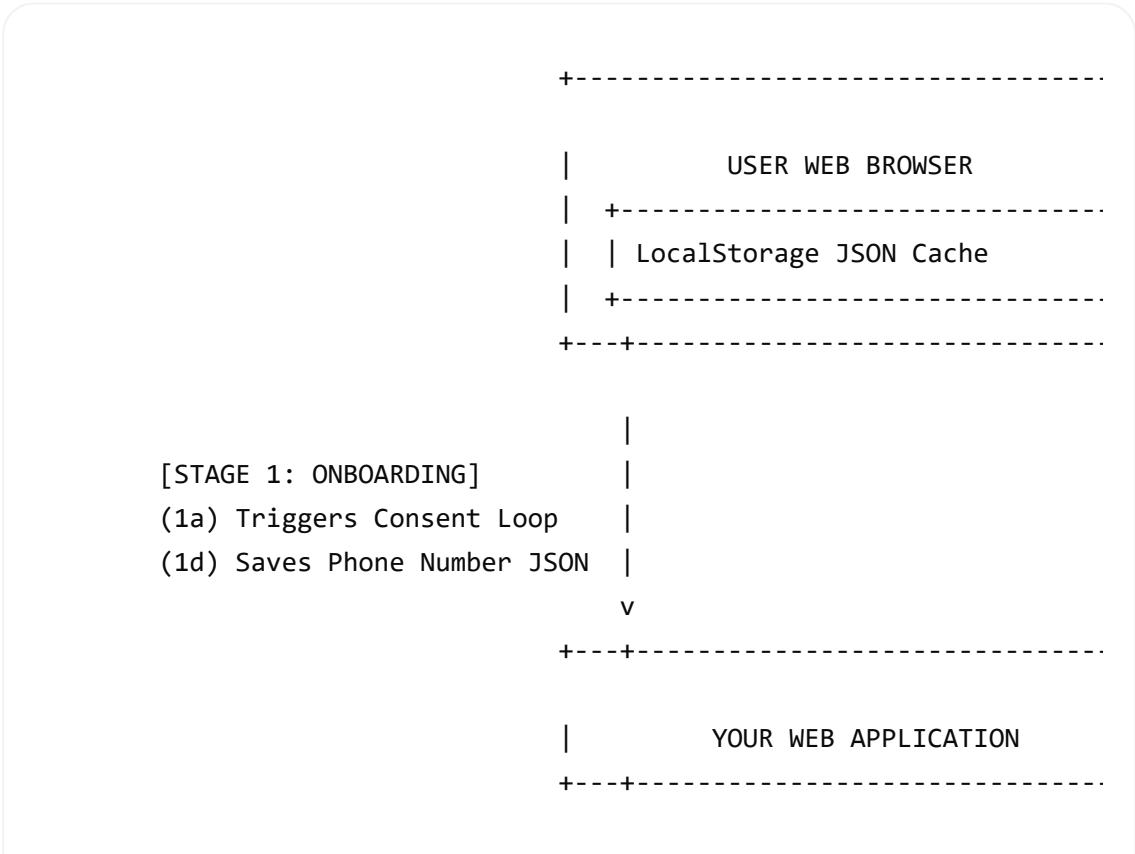
for my proof of concept (POC) data will stored in
localStorage using JSON. No database is required

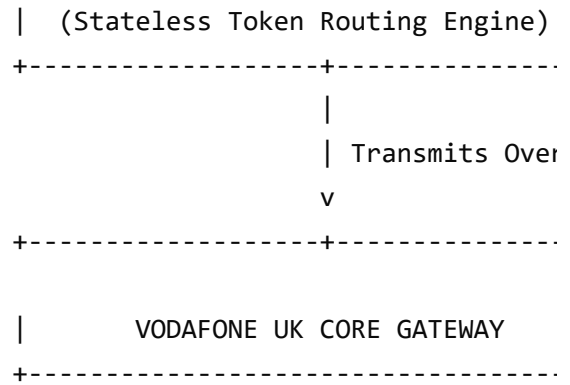
15:06

That keeps the PoC architecture remarkably lightweight and eliminates backend storage dependencies entirely. By leveraging `localStorage`, the user's browser acts as the single source of truth for their identity state.

Here is your updated, database-free data flow architecture designed specifically for this client-side configuration.

LocalStorage PoC Data Flow Architecture





Stage 1: Onboarding Flow (`localStorage` Write)

The browser drives the authorization loop, and your stateless backend acts as a proxy to handle the API credential handshake.

1. **User clicks "Onboard"** → Web app initiates Vodafone's 3-legged OAuth consent loop.
2. **User grants permission** → Vodafone redirects the browser back to your app with an Authorization Code.
3. **Token Exchange** → Web app sends the code to your backend, which swaps it for a temporary Vodafone access token.
4. **Data Retrieval (`GET`)** → Backend calls `GET /device-phone-number` , receives the phone number string, and passes it to the web app.
5. **Local Storage Write** → The front-end web app stringifies the data and saves it locally using JavaScript:

javascript

```
const msisdnData= { phoneNumber: "+447700900077", verifiedAt: Dat
localStorage.setItem('vodafone_poc_user', JSON.stringify(msisdnDat
```

Use code with caution.



javascript

```
const cachedUser = JSON.stringify(localStorage.getItem('vodafone_r
```

Use code with caution.



2. **Silent Handshake Request** → Web app passes `cachedUser.phoneNumber` directly in the payload to your backend.
3. **Execution (POST)** → Backend utilizes its system-wide 2-legged M2M token to call `POST /verify` silently against Vodafone's gateway.
4. **Instant Login** → Vodafone confirms the active device SIM matches the payload (`verified: true`). The web app updates the UI to logged-in status with zero user interaction. [↗](#)

PoC Configuration Constraints

- **Device Mapping:** This setup assumes the user returns using the exact same device and browser profile. Clearing browser cookies or cache will wipe the `localStorage` JSON string, resetting them to Stage 1.
- **Stateless Backend:** Your backend requires no state persistence, no schemas, and no file systems. It only needs functions to handle the authorization token trades and proxy the outward CAMARA endpoints. [↗](#)

Now that the architecture utilizes purely client-side local cache, would you like to design the **JSON data structure template** for the browser storage, or should we prepare the **frontend JavaScript fetch routines** for both paths?

AI responses may include mistakes. [Learn more](#)



lets do both

15:10

1. LocalStorage JSON Data Structure Template

To keep your Proof of Concept (PoC) organized and easy to debug, store your data in `localStorage` under a single, structured key (e.g., `'vodafone_poc_session'`).

This schema tracks both the user's phone identity and the verification state machine:

json

```
{
  "user": {
    "phoneNumber": "+447700900077",
    "countryCode": "GB"
  },
  "meta": {
    "onboardedAt": 1785251400000,
```



- **meta.stage State Machine Values:**
 - "EMPTY" : Default state. No data found (Triggers Stage 1 Onboarding).
 - "PENDING_CONSENT" : Redirected to Vodafone; waiting for the OAuth callback code.
 - "VERIFIED" : Active authenticated state (Triggers Stage 2 Silent Handshake on next visit).

2. Frontend JavaScript Implementation

This complete, modular frontend implementation coordinates the two execution stages. It reads from `localStorage`, handles the network handshakes, and communicates with your stateless backend proxy endpoints (`/api/onboard` and `/api/silent-verify`).

javascript

```
// Configuration Constants
const STORAGE_KEY = 'vodafone_poc_session';
const BACKEND_URL = 'https://yourpocdomain.local'; // Replace with your domain

/**
 * Core Orchestrator running immediately on page load
 */
async function initializePocWorkflow() {
  const session = getLocalSession();

  // If no data exists, the user must onboard via Stage 1 (GET Flow)
  if (!session || session.meta.stage === 'EMPTY') {
    console.log("No profile found. Ready for Stage 1 Onboarding.");
    updateUiState('onboarding_ready');
    return;
  }

  // If a profile exists, silently authenticate via Stage 2 (POST Flow)
  if (session.meta.stage === 'VERIFIED') {
```



```

* STAGE 1: Kickstart Onboarding
* Triggers the 3-Legged OAuth Consent Loop via your backend proxy
*/
function startOnboardingWorkflow() {
  updateLocalSession({ meta: { stage: 'PENDING_CONSENT' } });

  // Redirect the browser straight to your backend route,
  // which builds the secure Vodafone OAuth URL and bounces the user to
  window.location.href = `${BACKEND_URL}/api/auth/vodafone-connect`;
}

/**
 * STAGE 1 CALLBACK: Processes the returned OAuth code from the URL query
 * Call this function if your URL string contains a '?code=xyz' parameter
 */
async function handleOnboardingCallback(authCode) {
  try {
    updateUiState('processing_onboarding');

    // Send the auth code to the backend to swap for a token and call
    const response = await fetch(`${BACKEND_URL}/api/onboard`, {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ code: authCode })
    });

    if (!response.ok) throw new Error('Onboarding network exchange failed');
    const data = await response.json(); // Expected: { phoneNumber: "07700 900000", ... }

    // Save newly extracted data straight into the local web sandbox
    const newSession = {
      user: { phoneNumber: data.phoneNumber, countryCode: 'GB' },
      meta: { onboardedAt: Date.now(), lastVerifiedAt: Date.now(), stage: 'ONBOARDING_COMPLETED' }
    };
    localStorage.setItem(STORAGE_KEY, JSON.stringify(newSession));

    updateUiState('authenticated', data.phoneNumber);
  } catch (error) {
    console.error("Onboarding failed:", error);
    updateUiState('error', 'Failed to retrieve phone number from SIM.');
  }
}

```



```

async function executeSilentHandshake(phoneNumber) {
  try {
    const response = await fetch(`${BACKEND_URL}/api/silent-verify`, {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ phoneNumber: phoneNumber })
    });

    if (!response.ok) throw new Error('Verification network call failed');
    const data = await response.json(); // Expected: { verified: true,

    if (data.verified === true) {
      // Re-stamp validation timing metrics
      const session = getLocalSession();
      session.meta.lastVerifiedAt = Date.now();
      localStorage.setItem(STORAGE_KEY, JSON.stringify(session));

      updateUiState('authenticated', phoneNumber);
    } else {
      // SIM card swap or wrong device detected
      throw new Error('SIM mismatch detected by network gateway');
    }
  } catch (error) {
    console.warn("Silent handshake failed:", error.message);
    updateUiState('wifi_or_fallback_prompt');
  }
}

/**
 * Helpers: LocalStorage CRUD Utilities
 */
function getLocalSession() {
  const data = localStorage.getItem(STORAGE_KEY);
  return data ? JSON.parse(data) : { meta: { stage: 'EMPTY' } };
}

function updateLocalSession(updatedFields) {
  const current = getLocalSession();
  const merged = { ...current, ...updatedFields };
  localStorage.setItem(STORAGE_KEY, JSON.stringify(merged));
}

```



```
console.log(`UI State transitioned to: [${state}]`, customParam);
}
```

Use code with caution.



Now that the front-end storage format and step-by-step logic scripts are complete, we can move directly to the **stateless backend proxy routes** (`/api/auth/vodafone-connect` , `/api/onboard` , and `/api/silent-verify`) to process the actual tokens. Would you like to build those endpoints next?

AI responses may include mistakes. [Learn more](#)



Yes, but I need to do them in Java from an Openfire plugin using embedded Jetty

15:11

To build these stateless proxy routes within an **Openfire plugin**, you will implement a standard `Servlet` and register it with Openfire's internal web container (which runs on **embedded Jetty**).

Openfire provides a specialized plugin web context infrastructure that handles this mapping cleanly.

1. Plugin Configuration Framework (`plugin.xml`)

Your Openfire plugin must first register the servlet so Jetty knows where to route incoming browser requests. Add this servlet mapping inside your plugin's `plugin.xml` configuration file:



```

<version>1.0.0</version>
<minServerVersion>4.7.0</minServerVersion>

<!-- Registers the Jetty Servlet Mapper -->
<servlets>
  <servlet>
    <name>VodafoneProxyServlet</name>
    <class>org.example.openfire.vodafone.VodafoneProxyServlet<
    <url-pattern>/vodafone-poc/*</url-pattern>
  </servlet>
</servlets>
</plugin>

```

Use code with caution.



Note: Due to Openfire's core security filters, this servlet will be exposed inside the plugin context path at: `https://<openfire-host>:9094/plugins/vodafonecamarapoc/vodafone-poc/`

2. The Java Embedded Jetty Servlet Implementation

This standalone Java class extends `HttpServlet`. It exposes the three core endpoints needed by your JavaScript frontend. It uses native Java network features (`HttpClient`) to interact with Vodafone's CAMARA v2.1.0 gateway without requiring third-party library installations.

java

```

package org.example.openfire.vodafone;

import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.IOException;

```



```

// CAMARA Gateway Sandbox URLs (Swap with Live endpoints for prod)
private static final String VODAFONE_AUTH_URL = "https://vodafone
private static final String VODAFONE_GET_URL = "https://vodafone
private static final String VODAFONE_POST_URL = "https://vodafone

// Application Credentials (Supplied by your Vodafone Developer Po
private static final String CLIENT_ID = "YOUR_VODAFONE_CLIENT_
private static final String CLIENT_SECRET = "YOUR_VODAFONE_CLIENT_
private static final String REDIRECT_URI = "https://yourdomain.l

private final HttpClient httpClient = HttpClient.newHttpClient();

@Override
protected void doGet(HttpServletRequest req, HttpServletResponse resp) {
    String pathInfo = req.getPathInfo();

    // ROUTE 1: Kickstart 3-Legged OAuth User Consent Redirect
    if ("/auth/vodafone-connect".equals(pathInfo)) {
        String oauthAuthorizeUrl = "https://vodafone.com" +
            "?response_type=code" +
            "&client_id=" + URLEncoder.encode(CLIENT_ID, StandardCharsets.UTF_8) +
            "&redirect_uri=" + URLEncoder.encode(REDIRECT_URI, StandardCharsets.UTF_8) +
            "&scope=number-verification-verify-read";

        resp.sendRedirect(oauthAuthorizeUrl);
        return;
    }

    resp.sendError(HttpServletResponse.SC_NOT_FOUND, "Endpoint not found");
}

@Override
protected void doPost(HttpServletRequest req, HttpServletResponse resp) {
    String pathInfo = req.getPathInfo();
    resp.setContentType("application/json");
    resp.setCharacterEncoding("UTF-8");

    try {
        // ROUTE 2: Complete Onboarding (Swap Auth Code -> Fetch l

```



```

        "&code=" + URLEncoder.encode(authorizationCode)
        "&redirect_uri=" + URLEncoder.encode(REDIRECT_
        "&client_id=" + CLIENT_ID +
        "&client_secret=" + CLIENT_SECRET;

    HttpRequest tokenRequest = HttpRequest.newBuilder()
        .uri(URI.create(VODAFONE_AUTH_URL))
        .header("Content-Type", "application/x-www-form-urlencoded")
        .POST(HttpRequest.BodyPublishers.ofString(tokenRequest.build()));

    HttpResponse<String> tokenResponse = httpClient.send(tokenRequest, HttpResponse.BodyHandlers.ofString());
    String accessToken = extractJsonKey(tokenResponse.body(), "access_token");

    // Step B: Query Vodafone GET endpoint using retrieved token
    HttpRequest getMsisdnRequest = HttpRequest.newBuilder()
        .uri(URI.create(VODAFONE_GET_URL))
        .header("Authorization", "Bearer " + accessToken)
        .GET()
        .build();

    HttpResponse<String> msisdnResponse = httpClient.send(getMsisdnRequest, HttpResponse.BodyHandlers.ofString());

    // Return payload string straight back to local web client
    resp.getWriter().write(msisdnResponse.body());
    return;
}

// ROUTE 3: Returning Silent Handshake via 2-Legged M2M client
if ("/silent-verify".equals(pathInfo)) {
    String body = new String(req.getInputStream().readAllBytes());
    String targetPhoneNumber = extractJsonKey(body, "phone_number");

    // Step A: Request 2-Legged Client Credentials Token
    String m2mPayload = "grant_type=client_credentials" +
        "&client_id=" + CLIENT_ID +
        "&client_secret=" + CLIENT_SECRET;

    HttpRequest m2mRequest = HttpRequest.newBuilder()
        .uri(URI.create(VODAFONE_AUTH_URL))

```



```

// Step B: Run POST Verification pass payload
String verifyJsonBody = "{\"phoneNumber\":\"" + target
HttpRequest verifyRequest = HttpRequest.newBuilder()
    .uri(URI.create(VODAFONE_POST_URL))
    .header("Authorization", "Bearer " + m2mToken)
    .header("Content-Type", "application/json")
    .POST(HttpRequest.BodyPublishers.ofString(verifyJsonBody))
    .build();

HttpResponse<String> verifyResponse = httpClient.send(verifyRequest,
    HttpResponse.BodyHandlers.ofString());

resp.getWriter().write(verifyResponse.body());
return;
}

} catch (Exception e) {
    resp.setStatus(HttpServletResponse.SC_INTERNAL_SERVER_ERROR);
    resp.getWriter().write("{\"error\":\"Internal proxy process failed\"}");
    return;
}

resp.sendError(HttpServletResponse.SC_NOT_FOUND);
}

/**
 * Minimalist JSON string parser loop optimized specifically for loop
 * to eliminate deep compilation dependencies inside custom Openf
 */
private String extractJsonKey(String json, String key) {
    String searchKey = "\"" + key + "\"";
    int startIdx = json.indexOf(searchKey);
    if (startIdx == -1) return "";
    startIdx += searchKey.length();

    // Advance cursor past blank styling zones or object character
    while (startIdx < json.length() && (json.charAt(startIdx) == ' ' || json.charAt(startIdx) == ','))
        startIdx++;
}

int endIdx = startIdx;

```



Use code with caution.



3. Openfire Lifecycle Activation

To hook the servlet lifecycle into Openfire when the server boots up, configure your primary `Plugin` lifecycle adapter class like this:

java

```
package org.example.openfire.vodafone;

import org.jivesoftware.openfire.container.Plugin;
import org.jivesoftware.openfire.container.PluginManager;
import java.io.File;

public class VodafonePocPlugin implements Plugin {

    @Override
    public void initializePlugin(PluginManager manager, File pluginDir) {
        // Openfire automates servlet registration from plugin.xml and
        System.out.println("Vodafone CAMARA Proxy Servlet successfully");
    }

    @Override
    public void destroyPlugin() {
        System.out.println("Vodafone CAMARA Proxy Servlet context torn");
    }
}
```

Use code with caution.



Now that both your frontend `localStorage` engine and backend Openfire Java servlet infrastructure are written, do you want to finalize the setup by reviewing how to handle **Cross-Origin Resource Sharing (CORS)** flags inside the servlet so your web application can safely talk to the Openfire Jetty port?



yes

15:12

To ensure your web application's frontend can successfully communicate with your Openfire embedded Jetty servlet across different ports or subdomains, you must handle Cross-Origin Resource Sharing (CORS).

Browsers will block these requests by default unless your Java servlet explicitly returns the correct HTTP handshake headers.

1. Handling the Preflight (OPTIONS) Request

When your frontend sends a `POST` request with JSON data (`Content-Type: application/json`), the browser first automatically transmits an `OPTIONS` request (known as a "preflight" check) to verify that the server allows the incoming cross-origin call. Your servlet must intercept this and reply with an immediate `200 OK` status and the appropriate access headers.

2. Updated Servlet with Native CORS Handling

Add an explicit `doOptions` method to your servlet and update the global header injection inside `doPost` . Here is how to modify your `VodafoneProxyServlet` class to safely enable cross-origin browser requests:

java

```
package org.example.openfire.vodafone;

import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.IOException;

public class VodafoneProxvServlet extends HttpServlet {
```



```
@Override
protected void doOptions(HttpServletRequest req, HttpServletResponse resp) {
    setCorsHeaders(resp);
    resp.setStatus(HttpServletResponse.SC_OK);
}

/**
 * Helper utility to inject required security policy headers onto
 */
private void setCorsHeaders(HttpServletResponse resp) {
    // Use ALLOWED_ORIGIN instead of "*" if your frontend passes it
    resp.setHeader("Access-Control-Allow-Origin", ALLOWED_ORIGIN);
    resp.setHeader("Access-Control-Allow-Methods", "POST, GET, OPTIONS");
    resp.setHeader("Access-Control-Allow-Headers", "Content-Type, Authorization");
    resp.setHeader("Access-Control-Max-Age", "3600"); // Caches the result
}

@Override
protected void doGet(HttpServletRequest req, HttpServletResponse resp) {
    // Inject CORS headers onto incoming GET paths
    setCorsHeaders(resp);

    String pathInfo = req.getPathInfo();
    if ("/auth/vodafone-connect".equals(pathInfo)) {
        // OAuth Redirect engine Logic here...
        return;
    }
    resp.sendError(HttpServletResponse.SC_NOT_FOUND);
}

@Override
protected void doPost(HttpServletRequest req, HttpServletResponse resp) {
    // ALWAYS inject CORS headers at the very beginning of process
    setCorsHeaders(resp);

    String pathInfo = req.getPathInfo();
    resp.setContentType("application/json");
    resp.setCharacterEncoding("UTF-8");

    try {
```



```
    }  
  } catch (Exception e) {  
    resp.setStatus(HttpServletResponse.SC_INTERNAL_SERVER_ERROR);  
    resp.getWriter().write("{\"error\":\"" + e.getMessage() + "  
    return;  
  }  
  resp.sendError(HttpServletResponse.SC_NOT_FOUND);  
}  
}
```

Use code with caution.

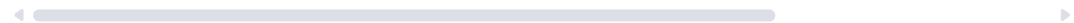


Important Jetty Port Reminder

By default, Openfire's web container binds unsecured plugins to port **9090** and TLS/SSL secured plugins to port **9094**. Because the CAMARA Number Verification 2.1.0 API deals with sensitive subscriber identities, your frontend script must point explicitly to the secure port:

javascript

```
// Ensure you target the secure HTTPS Jetty port inside your JavaScript  
const BACKEND_URL = 'https://<your-openfire-host>:9094/plugins/vodafo
```



Use code with caution.



Now that your end-to-end sandbox stack is complete—spanning frontend `localStorage`, a stateless Openfire Java servlet layout, and matching embedded Jetty CORS handling—do you need any final configuration details regarding **compilation dependencies** inside your plugin project, or are you fully equipped to start building your Proof of Concept?

AI responses may include mistakes. [Learn more](#)



probably a template HTML front-end web page to server
from my plugin and host the front-end javascript and API ...

15:15

To keep your Proof of Concept self-contained, you can serve the frontend directly from your Openfire plugin. Embedded Jetty looks for web assets inside a folder named `web` in your plugin's source directory structure.

Here is the setup to serve a clean, single-page dashboard.

1. File Structure Layout

Place your web files within your plugin development folder like this:

text

```
your-plugin-root/  
|  
├─ src/java/org/example/openfire/... (Your Java Servlet)  
├─ web/  
|   ├─ index.html      <-- Create this file  
|   └─ poc-client.js    <-- Put the Frontend JavaScript code here  
└─ plugin.xml
```

Use code with caution.



Note: Openfire's Jetty container automatically serves files in the `web` folder statically. You can access this page at: `https://<openfire-host>:9094/plugins/vodafonecamarapoc/index.html`

2. The Template HTML Page (`index.html`)

This responsive HTML dashboard serves as the visual interface for your PoC. It uses distinct layout cards that toggle dynamically based on the user's state.



```
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>Vodafone CAMARA v2.1.0 PoC Dashboard</title>
<style>
  :root {
    --voda-red: #E60000;
    --dark-gray: #333333;
    --light-bg: #F4F4F4;
  }
  body {
    font-family: -apple-system, BlinkMacSystemFont, "Segoe UI",
    sans-serif;
    background-color: var(--light-bg);
    color: var(--dark-gray);
    margin: 0;
    padding: 40px 20px;
    display: flex;
    justify-content: center;
  }
  .container {
    max-width: 500px;
    width: 100%;
    background: #FFFFFF;
    padding: 30px;
    border-radius: 8px;
    box-shadow: 0 4px 12px rgba(0,0,0,0.1);
  }
  .header {
    text-align: center;
    border-bottom: 2px solid var(--light-bg);
    padding-bottom: 20px;
    margin-bottom: 25px;
  }
  .logo {
    font-size: 24px;
    font-weight: bold;
    color: var(--voda-red);
  }
  .status-badge {
    display: inline-block;
    padding: 6px 12px;
    border-radius: 20px;
    background-color: var(--light-bg);
    color: var(--dark-gray);
    font-weight: bold;
  }
```



```
text-align: center;
margin-top: 20px;
}
.card.active {
  display: block;
}
.btn {
  background-color: var(--voda-red);
  color: white;
  border: none;
  padding: 12px 24px;
  font-size: 16px;
  font-weight: bold;
  border-radius: 4px;
  cursor: pointer;
  width: 100%;
  transition: background 0.2s;
}
.btn:hover {
  background-color: #B30000;
}
.data-box {
  background: var(--light-bg);
  padding: 15px;
  border-radius: 4px;
  text-align: left;
  font-family: monospace;
  font-size: 13px;
  overflow-x: auto;
  margin-top: 15px;
}
.alert-warning {
  background-color: #FFF3CD;
  color: #856404;
  padding: 15px;
  border-radius: 4px;
  border: 1px solid #FFEBAE;
  text-align: left;
}
</style>
```



```

</div>

<!-- STAGE 1 CARD: Brand New User Onboarding -->
<div id="card-onboarding_ready" class="card">
  <h3>Welcome to Onboarding</h3>
  <p>Register securely using your active mobile network profile
  <button class="btn" onclick="startOnboardingWorkflow()">Onboa
</div>

<!-- PROCESSING CARD: Token / Network Exchange Pending -->
<div id="card-processing" class="card">
  <h3>Network Exchange in Progress</h3>
  <p>Talking to Vodafone Core API Infrastructure... Please do not
</div>

<!-- STAGE 2 CARD: Returning Session Verified Silently -->
<div id="card-authenticated" class="card">
  <h3 style="color: green;">✓ Authenticated Silently</h3>
  <p>Welcome back! You have been logged in automatically via a token
  <div class="data-box">
    <strong>Active Identity Payload:</strong>
    <pre id="json-display"></pre>
  </div>
  <br>
  <button class="btn" style="background:#555;" onclick="clearPayload()">Clear
</div>

<!-- WI-FI / COVERAGE FALLBACK PROMPT -->
<div id="card-fallback" class="card">
  <h3>Network Verification Required</h3>
  <div class="alert-warning">
    <strong>Silent Handshake Failed:</strong> We could not establish a connection
  </div>
  <p>To verify instantly, please temporarily <strong>turn off your mobile data</strong>
  <button class="btn" onclick="retrySilentVerification()">Retry
</div>
</div>

<!-- Hook up your externalised JavaScript Logic script -->
<script src="poc-client.js"></script>

```




```

const urlParams = new URLSearchParams(window.location.search);
const authCode = urlParams.get('code');

if (authCode) {
    handleOnboardingCallback(authCode);
} else {
    initializePocWorkflow();
}
});

// Extends the base script to bridge the visual card state engine
function updateUiState(state, customParam = '') {
    const badge = document.getElementById('app-status');

    // Remove active state from all wrapper layout elements
    document.querySelectorAll('.card').forEach(card => card.classList

switch(state) {
    case 'onboarding_ready':
        badge.innerText = "Guest Mode";
        badge.style.background = "#E0E0E0";
        document.getElementById('card-onboarding_ready').class:
        break;
    case 'processing_onboarding':
    case 'verifying_silently':
        badge.innerText = "Verifying SIM...";
        badge.style.background = "#FFEBAA";
        document.getElementById('card-processing').classList.:
        break;
    case 'authenticated':
        badge.innerText = "SIM Verified";
        badge.style.background = "#D4EDDA";
        badge.style.color = "#155724";
        document.getElementById('json-display').innerText = J;
        document.getElementById('card-authenticated').classLi:
        break;
    case 'wifi_or_fallback_prompt':
        badge.innerText = "Action Required";
        badge.style.background = "#F8D7DA";
        badge.style.color = "#721C24";

```



```
if (session && session.user && session.user.phoneNumber) {  
    executeSilentHandshake(session.user.phoneNumber);  
}  
  
function clearPocSession() {  
    localStorage.removeItem('vodafone_poc_session');  
    window.location.href = window.location.pathname; // Reload wit  
}  
</script>  
</body>  
</html>
```

Use code with caution.



3. Verification Steps

1. Package your plugin into a `.jar` file and drop it into Openfire's `plugins/` directory.
2. Navigate your mobile browser to `https://<your-openfire-ip>:9094/plugins/vodafonecamarapoc/index.html`.
3. Tap **Onboard with Vodafone** to verify the end-to-end user consent loop, `GET` retrieval data extraction, local cache persistence, and subsequent silent `POST` checks.

You now have a complete, functional blueprint for your PoC. Do you need any assistance with the compilation build commands or testing parameters, or are you ready to launch the system?

AI responses may include mistakes. [Learn more](#)



To package your Openfire plugin with the correct dependencies, you need a standard Maven `pom.xml` configuration file. [↗](#)

Openfire development requires the `openfire-server` dependency provided by Ignition Realtime. Crucially, the **Jakarta Servlet API** must be marked as `<scope>provided</scope>` because Openfire's embedded Jetty container already provides it at runtime. This avoids dependency version conflicts when compiling. [↗](#)

Here is the production-ready `pom.xml` configuration file for your project:

xml

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://apache.org"
  xmlns:xsi="http://w3.org"
  xsi:schemaLocation="http://apache.org http://apache.org">
  <modelVersion>4.0.0</modelVersion>

  <groupId>org.example.openfire.vodafone</groupId>
  <artifactId>vodafone-camara-poc</artifactId>
  <version>1.0.0</version>
  <packaging>jar</packaging>

  <name>Vodafone CAMARA PoC Plugin</name>
  <description>Openfire plugin handling Vodafone UK CAMARA APIs via

  <properties>
    <maven.compiler.source>17</maven.compiler.source>
    <maven.compiler.target>17</maven.compiler.target>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
    <!-- Align this version with your targeted Openfire environment -->
    <openfire.version>4.9.0</openfire.version>
  </properties>

  <!-- Configure the Ignition Realtime Repository to fetch official
  <repositories>
    <repository>
      <id>igniterealtime</id>
```



```

<dependency>
  <groupId>org.igniterealtime.openfire</groupId>
  <artifactId>xmppserver</artifactId>
  <version>${openfire.version}</version>
  <scope>provided</scope>
</dependency>

<!-- Jakarta Servlet API required for standard modern Openfire
<dependency>
  <groupId>jakarta.servlet</groupId>
  <artifactId>jakarta.servlet-api</artifactId>
  <version>6.0.0</version>
  <scope>provided</scope>
</dependency>
</dependencies>

<build>
  <finalName>vodafonecamarapoc</finalName>
  <sourceDirectory>src/java</sourceDirectory>

  <resources>
    <!-- Instructs Maven to include plugin.xml and the static
    <resource>
      <directory>src/resources</directory>
      <includes>
        <include>plugin.xml</include>
      </includes>
    </resource>
    <resource>
      <directory>web</directory>
      <targetPath>web</targetPath>
    </resource>
  </resources>

  <plugins>
    <!-- Compiler Configuration Plugin -->
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-compiler-plugin</artifactId>
      <version>3.11.0</version>

```



```
<plugin>
  <groupId>org.apache.maven.plugins</groupId>
  <artifactId>maven-jar-plugin</artifactId>
  <version>3.3.0</version>
  <configuration>
    <archive>
      <manifestEntries>
        <Plugin-Class>org.example.openfire.vodafoi
      </manifestEntries>
    </archive>
  </configuration>
</plugin>
</plugins>
</build>
</project>
```

Use code with caution.



Folder Structure Adjustment for this POM

To match the standard Maven layout used above, ensure your root folder is organized exactly like this before running compilation commands:

text

```
your-project-root/
|
├─ pom.xml
├─ web/
|   ├─ index.html
|   └─ poc-client.js
└─ src/
    ├─ java/
    |   └─ org/example/openfire/vodafone/
    |       ├─ VodafonePocPlugin.java
    |       └─ VodafoneProxyServlet.java
    └─ resources/
```



bash`mvn clean package`

Use code with caution.



This command compiles your code and outputs `vodafonecamarapoc.jar` inside your newly created `target/` directory, which is ready to drop straight into Openfire.

Do you need help with **how to deploy the compiled JAR file** into Openfire or configure any **SSL settings** for port 9094 to run your live mobile tests?

AI responses may include mistakes. [Learn more](#)



I think i am good now

15:19



Good luck building out and validating your Vodafone UK CAMARA API system. If you hit any network edge cases, certificate matching hurdles, or token expiration issues once you transition to live testing, feel free to drop back in to troubleshoot. Happy coding!

AI responses may include mistakes. [Learn more](#)

